Research Gaps in AI-ECG for Heart Failure Diagnosis

Prepared 29 October 2026 from three sources: an official health-technology brief, a peer-reviewed narrative review and a specialised news report.

Summary

The main gaps in AI-based ECG analysis for heart failure (HF) diagnosis are weak external validation, little proof of clinical benefit, low positive predictive value, unequal performance across patient groups and unresolved implementation questions. Each gap below is supported by at least one of the three sources listed later in this document.

Main research gaps

1. Limited external and prospective validation

The 2026 narrative review found that most studies were retrospective and single-center, often validated only internally. External and prospective evaluation was uncommon, and reporting of blinding, reference standards and diagnostic timelines was frequently incomplete.

2. Accuracy has not been shown to improve outcomes

The Singapore government brief reports that evidence for clinical utility remains limited. In the EAGLE trial, Anumana's tool changed detection of low ejection fraction only slightly (0.9% vs 0.6%, not statistically significant) and did not significantly change guideline-directed therapy. Effects on HF hospitalisation, mortality and long-term management are largely unknown.

3. Low positive predictive value

Against echocardiography, sensitivity and specificity were moderate to good, but positive predictive value ranged only from 16.4% to 43.3%. The trade-off between false positives and added echocardiography demand is poorly studied. The news report shows a similar pattern: about two-thirds of suspected HF cases flagged by the AI stethoscope were later ruled out.

4. Fairness and generalisability

Model performance varied across institutions and demographic subgroups, and only a minority of studies used explainability techniques. The official brief also notes that Asian patients were under-represented in the evidence base (0% to 19.9% of participants), which limits applicability to local populations.

5. Real-world adoption and workflow

In the TRICORDER trial, researchers said wider rollout would need better integration into GPs' routines, because many surgeries stopped using the device after a year. The official brief adds open questions about clinician trust, clinical governance, cybersecurity and managing a possible rise in echocardiography referrals.

6. Economic evidence and unreviewed products

Formal cost-effectiveness analyses were uncommon across the studies in the narrative review. In the official brief, two of the four FDA-cleared tools (Tempus ECG-Low EF and Bunkerhill ECG-EF) had no peer-reviewed publications, only data submitted to the FDA.

7. Narrow focus and no tracking over time

The narrative review recommends that future work use standardized reporting frameworks such as TRIPOD-AI and evaluate patient-centred endpoints, not only discrimination metrics. Local clinicians quoted in the official brief also noted that current approved tools give threshold cut-offs and may not be useful for monitoring improvement or worsening over time.

The three sources

Official source

- **Reference:** Agency for Care Effectiveness (Singapore Ministry of Health). *Artificial intelligence-enabled electrocardiogram (AI-ECG) to detect reduced left ventricular ejection fraction.* ACE Brief HSB-M 01/2026, December 2025.
- **Link:** https://isomer-user-content.by.gov.sg/68/628aff49-ab5c-4ffb-93f2-ff76f4721da6/Artificial%20intelligence-enabled%20electrocardiogram%20(AI-ECG)%20to%20detect%20reduced%20left%20ventricular%20ej.pdf
- **Used for:** gaps 2, 3, 4, 5, 6 and 7.

Scientific article

- **Reference:** Chepkunov D, Chung D, Ghanbari K, Khanna B, Yakubov N, Babu M, Babu B. *Artificial intelligence-augmented electrocardiogram for heart failure diagnosis and risk prediction: a narrative review.* Journal of Medical Artificial Intelligence 2026;9:36. doi:10.21037/jmai-2025-178
- **Link:** https://jmai.amegroups.org/article/view/11269/html
- **Used for:** gaps 1, 4, 6 and 7.

Newsletter / specialised news

- **Reference:** HospiMedica International staff writers. *AI Stethoscope Detects Three Heart Conditions In 15 Seconds.* 4 September 2025.
- **Link:** https://www.hospimedica.com/artificial-intelligence/articles/294806514/ai-stethoscope-detects-three-heart-conditions-in-15-seconds.html
- **Used for:** gaps 3 and 5.

Cautions before citing

- The ACE brief states that it rests on a limited literature search and does not critically appraise the evidence.
- The narrative review applied no formal risk-of-bias assessment, and its screening was done by student authors.
- The HospiMedica piece covers an AI stethoscope (single-lead ECG plus heart sounds), not a 12-lead ECG. Its figures also differ from other outlets: it reports 96 clinics, while other reports say 205 practices.